GAD Lectures

Lecture 1:

Introduction of Unity.
What is difference between UI and UX in Unity 3D.

Lecture 2:

1. Search on Google => unity hub download.
2. Click => https://unity.com/download (Download for windows).
3. Then installation .

Lecture #3:

1. How to add the footpath texture
2. DOWNLOAD footpath texture jpg
3. Import in the assets
4. Select the terrain
5. Create new layer with that
6. Paint it on the footpath

Lecture 4:

- ➔ How to use the trees:
- ➔ drag the downloaded pkg and drag and drop that in the project
- ➔ Import all dependencies
- ➔ Folder will start to show you.
- ➔ click on the things that started to show the preview these are prefab that is built in. we have to pick and use that.
- ➔ select that tree and drop on the terrain. Can also increase and decrease the size of tree.
- ➔ position after click and select that in hierarchy and using scale

- ➔ select ground => Trees icons => Edit tress => add trees => assign trees (prefab) select the tree from project and drop it in Add Tree.
- ➔ select Brush size 60/or more and density 60(number of trees)
- ➔ To remove specific tree => shift + click

jider camera laa k jaanaa chate ho phly oss chiz ko view ma laa ao, phir camera per click kar ka Press **Ctrl + shift + f.**

Lecture 5 & 6:

- Tool option is started to show in the top.
- openfolder => models =>prefeb => drag and drop

- models: jis per koi work naa huva ho
- prefeb: jis per phly saa e sara work huva ho

- to move the camera along with the car you have to make the main camera child of the model. drag the main camera and paste in the prefeb folder

How to add the controller

- Selcet the model in Hierarchy => tools => bonecracker => relastic car controller => add main controller t vehicles.
- If errors check kry k error kis chiz ka ha jaise k wheels ka han to aap model open kry gaa fir wheels dhondy ga and free oss ko wha saa drag and drop kr daa.
- Body colidator => select body => add component => box colider

boundry walls lagna b zarrori han hum cube lagty han iss ma phly e box colidator laga hota han . Mesh renderd iss ma ak option hota han oss ko untick krny saa show hota han k jaise dosri side pr envirnoment han lakin cross ni kry gaa.

<u>Lecture 7 & 8:</u>

CREATE MODEL:

How to Import a RCC Package?

- First Download RCC Package.
- Copy RCC Package then Paste in Assets Folder in Unity.

How to Train a RCC Model (Car Model)?

- Open **assets** in -> **Realistic Car Controller,**
- Then open -> **Model** folder
- Then open -> **M3_E36** folder
- Select -> **M3_E36** Car
- Then open:
    - ➔ **Tools -> BoneCrakerGame -> Realistic Car Controller -> Add main Controller to vehicle**
- Click on **Wheel/Tiar**
- Apply all **Wheels**
    - **Wheels Left-Right**
    - **Wheels Left-Left**
    - **Wheels Right/Rear-Rear**
    - **Wheels Right/Rear-Left**
- Click on **Create Wheel Collider**
- Then Apply on **Body**
    - ➔ **Box Collider**
- Then **COM**:
    - ➔ Back arrow min
    - ➔ Down arrow min

What is Script in Unity Hub?

In Unity Hub, a "Script" typically refers to a type of file used in Unity projects to create and define behaviors, interactions, and functionalities within the game or application. These scripts are written in programming languages like C#, UnityScript (also known as JavaScript for Unity), or Boo.

How to Create a New Script?

- In the Unity Editor, navigate to the "**Assets**" folder or the folder where you want to create the script.
- Right-click within the folder or empty space in the "**Project**" window.
- Hover over "**Create**" and then select the appropriate scripting language for your new script (e.g., "**C# Script**").
- File Name -> **Camera Setting**
- Click on Script file and Open in Visual Studio
- Script Name and ClassName same.
- Script attaches with car model

Explanation of the Five Functions:

1. **Invoke Function:** Calls a specified function after a specified delay or repeatedly at set intervals.
2. **Start Function:** Executes once when an object is enabled, used for initialization.
3. **Update Function:** Executes every frame, commonly used for continuous tasks like input handling.
4. **FixedUpdate Function**: Runs at fixed intervals synchronized with the physics engine, suitable for physics-related calculations.
5. **LateUpdate Function:** Executes after Update, often used for tasks dependent on other updates, like camera manipulation.

Code Camera Set:

```
using UnityEngine;
public class Camera_Setting : MonoBehaviour
{
   public float x=0;
    public float y;
     public float z=0;
     public GameObject car;

void update(){
     y = car.transform.eulerAngles.y;
     transform.eulerAngles= new Vector3(x,y,z);
```

}
}

Important Note:

- **Camera Set:**
    - Control + Shift + F
- **RCCCamera script**

  Player vehicle - car apply(M3_E36)
    - collect car transform rotation
    - eulerAngles use for Collect All Rotations

Lecture 9 & 10:

How to install Ai Navigation on Unity?

1. Click on "**Window**" in the top bar.
2. Then select "**Package Manager**".
3. Under "**Packages: Unity Registry"** find and install "**AI Navigation**".
4. Once installed, you will see an **AI** option in the **Window** menu.

Now, let's integrate AI Navigation with an RCC Prefab Car:

1. Choose the "**M3_E36 Car**" from the available options.
2. Navigate to:
    - **Tools -> BoneCrakerGame -> Realistic Car Controller -> AI Controller -> Add the AI Controller to the vehicle.**
3. In the Inspector tab of the RCC AI Car Controller:
    - Select the desired AI Type:
        - **Follow Waypoints**
        - Chase Target
        - Follow target

To set up AI Waypoints in your scene:

1. Select the "**M3_E36 Car**"
2. Open:
    - **Tools -> BoneCrakerGame -> Realistic Car Controller -> AI Controller -> Add AI Waypoints to your scene.**
3. Create Waypoints by pressing Shift and clicking on the road to define the car's direction (similar to creating a road system).

Important Note:

- And always use your car model not prefab.
- Ai Car Speed increase and decrease.

Baking AI Navigation in Unity

- Select "**Static**" for the **terrain**.
- Navigate to the **Window**: -> Click on "**AI**."

- Choose "**Navigation (Obsolete)**".
- Show the Navigation (Obsolete) menu with Inspector:
  | Agents | Areas | **Bake** | Object |
- Click on "**Bake**" to initiate the process.

Create a simple Script for coins collection system in Unity using C#:

```
using UnityEngine;
public class Coin : MonoBehaviour
{
   public GameObject coin1, coin2;
void Start(){ }
void Update(){ }

   private void OnTriggerEnter(Collider other)
   {
      coin1.SetActive(false);
      coin2.SetActive(true);
   }
}
```

Mid

Lecture 11 & 12:

How to Design UI in Unity?

- Design the canvas using elements like panels, buttons, text fields, and images.
- Use the Hierarchy window to organize UI elements.

Generate a New Scene

Navigate to the **File** menu, select **New Scene**, and proceed to **Create** Scene.

Creating a Canvas with Panels, Buttons, Images, and Text:

- **Automatically Create a Canvas:**
  - Creating a new panel will automatically generate a Canvas.
  - Right-click in the Hierarchy window.
  - Hover over "UI" and select "Panel."
  - Right-click on the canvas in the Hierarchy window.
  - Create additional UI elements like Buttons, Images, and Text as needed.

Scene 1: Main Home

Step 1: Generate Two Panels:

- MainPanel
    - Play
    - Privacy
    - More
    - Exit
- ExitPanel
    - Yes
    - No

Once the two panels are created, assign the onClick functions for each button within the respective panels.

Step 2: Establish 'ScriptHolder':

- Generate a new object named 'ScriptHolder'.
- Apply the 'Main_UI' script to the 'ScriptHolder'.
- Implement the application of 'ExitPanel' within the 'Main_UI' script.

Scene 2: Level Selection

Step 1: Generate One Panels:

- LevelsPanel
    - Level1
    - Level2
    - Level3
    - Level4
    - Level5
    - Home
    - Back

Once the Created a panel, Then assign the onClick functions for each button within the respective panel.

Step 2: Establish 'ScriptHolder':

- Generate a new object named 'ScriptHolder'.
- Apply the 'Main_UI' script to the 'ScriptHolder'.

Scene 3: Level 1

Step 1:

- First Create Complete a work Of mid in which use like, terrain AiCar, MyCar etc.

Step 2: Generate Panels:

Then Apply a **RCC Canvas** from RCC folder.
Then, Create **Pause Button** in Canvas.
Then Create Three Panels:

1- Pause Panel
- Resume
- Restart
- Home

2- Complete Panel
- Next Level
- Restart
- Home

3- Fail Panel
- Restart
- Home

Once the Created a panel, Then assign the onClick functions for each button within the respective panel.

Step 3: Establish 'ScriptHolder':

- Generate a new object named 'ScriptHolder'.
- Apply the ' Level_Com_Fail' script to the 'ScriptHolder'.
- Implement the application of Three Panels 'PausePanel, CompletePanel, FailPanel' within the 'Level_Com_Fail' script.

Step 4: How to Create a Finish Line?

- Create **Finish Line** (Sphere).
- Then apply ‘Game_play Script.
- Finish Line **is Trigger** and Also **Mesh Render** Off

Apply Tags On Cars

- Then apply **‘Player’** Tag on **MyCar** and Car Body.
- Then apply **‘Enemy’** Tag on **AICar** and Car Body.

APK Build:

First install SDK, NDK and JDK Tools

Build Setting:

Navigate to the **File** menu, select **Build Settings** and proceed to **Scene in Build**.
Next, Choose Platform:

- Windows
- **Android**

After selecting Android, **switch** the platform. (Please wait for approximately 10 minutes.)
Then Click **Player Settings** and navigate to the Player Options.
Follow these two steps:

- Modify **Default Icons**.
- Specify **Resolution and Presentation** settings.

  Finally, choose the desired orientation:

  - Select "**Landscape Left**."

Then **Build.** (Please wait for approximately 15 minutes.)